



Software Architecture: Principles and Practices

Software Requirements Specification (SRS)

Community Tool Lending Platform

Version 1.0

Prepared by:

Mohit Yash P – 2024BCD0026

Vijaya Sai – 2024BCS322

Harish – 2024BCS318

Guru Charan – 2024BCS334

Submitted To:

Dr. Athira B

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Revision History

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the **Community Tool Lending Platform (CTLP)**. This document serves as a reference for the development team throughout the software development lifecycle and ensures a common understanding of the project's objectives, scope, and expected functionality before implementation begins.

1.2 Scope

The Community Tool Lending Platform (CTLP) is a web-based application that enables users to lend and borrow tools within a community in a secure and organized manner. The platform allows owners to list tools, borrowers to search and request available tools, and administrators to manage users and platform activities.

The first version of the system focuses on the core functionalities, including user authentication, tool management, reservation handling, lending agreements, in-app notifications, reviews and ratings, and administrative controls. The system aims to simplify community-based tool sharing while maintaining transparency and accountability between users.

1.3 Definitions, Acronyms and Abbreviations

Term	Description
CTLP	Community Tool Lending Platform
JWT	JSON Web Token used for secure user authentication
Admin	System administrator responsible for managing users and platform activities
Borrower	Registered user who borrows tools from other users
Owner	Registered user who lists and lends tools
SRS	Software Requirements Specification
UI	User Interface
API	Application Programming Interface

1.4 References

The following resources are considered during the preparation and development of this project:

- IEEE Software Requirements Specification (IEEE 830/ISO 29148) Guidelines
 - Documentations of the Technologies used (JWT, React, etc..)
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1.5 Document Overview

This document presents the software requirements for the Community Tool Lending Platform. It includes the overall system description, stakeholder identification, user characteristics, functional requirements, non-functional requirements, external interface requirements, business rules, assumptions,

and constraints. These requirements provide the foundation for the subsequent system design, implementation, testing, and deployment phases of the project.

2. Overall Description

2.1 Product Perspective

The Community Tool Lending Platform (CTLP) is a standalone web application designed to facilitate community-based tool sharing. The platform provides a centralized system where users can list tools for lending, browse available tools, request borrowings, and manage reservations in a secure and organized manner.

The application consists of three primary user roles: Borrower, Owner, and Administrator. Each role has specific responsibilities and permissions to ensure smooth operation and proper access control. The first version of the system focuses on delivering the essential functionalities required for an efficient and reliable tool lending platform.

2.2 Product Functions

The major functions provided by the system include:

- User registration and secure authentication.
- Role-based access control for Borrowers, Owners, and Administrators.
- Tool listing, updating, and removal.
- Searching and browsing available tools.
- Borrow request and approval management.
- Reservation tracking and return management.

- Digital lending agreement between the owner and borrower.
 - In-app notifications for important activities.
 - Reviews and ratings for completed borrowings.
 - Administrative management of users, listings, and platform activities.
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2.3 Stakeholders

The primary stakeholders of the Community Tool Lending Platform are:

Borrowers

Users who browse, request, borrow, and return tools available on the platform.

Owners

Users who list their tools, manage availability, approve or reject borrowing requests, and monitor lending history.

Administrators

Users responsible for maintaining the platform by managing users, monitoring listings, resolving disputes, and overseeing overall system activities.

2.4 User Characteristics

Borrower

- Basic knowledge of using a web application.
- Can search for available tools.
- Can submit borrowing requests.
- Can return borrowed tools and provide reviews.

Owner

- Basic knowledge of using a web application.
- Can manage personal tool listings.
- Can approve or reject borrowing requests.
- Can maintain tool availability and lending records.

Administrator

- Has complete access to platform management features.
 - Can manage users and tool listings.
 - Can resolve disputes and monitor platform statistics.
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2.5 Operating Environment

The system will operate as a web-based application accessible through modern web browsers.

Software Environment

- Frontend: React
- Backend: Node.js with Express.js
- Database: MySQL

Client Requirements

- Modern web browser (Google Chrome, Microsoft Edge, Mozilla Firefox, etc.)
 - Stable internet connection.
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2.6 Design Constraints

The development of the system is subject to the following constraints:

- The project follows the Waterfall Software Development Life Cycle.

- The application is developed as a web-based platform.
 - JWT is used for user authentication.
 - Images are stored locally.
 - Notifications are limited to in-app notifications in Version 1.
 - The system supports only the core functionalities defined within the project scope.
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2.7 Assumptions and Dependencies

The following assumptions are considered during the development of the system:

- Users provide valid registration information.
 - All users have access to an internet connection and a supported web browser.
 - Owners are responsible for maintaining the condition of their listed tools.
 - Borrowers agree to the lending agreement before borrowing a tool.
 - Future enhancements such as email notifications, cloud storage, and online payments are outside the scope of Version 1.
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3. Specific Requirements

3.1 Functional Requirements

The Community Tool Lending Platform shall provide the following functionalities.

User Management

FR-1: The system shall allow new users to register using their personal details.

FR-2: The system shall authenticate registered users using secure login credentials.

FR-3: The system shall provide role-based access for Borrowers, Owners, and Administrators.

FR-4: Users shall be able to view and update their profile information.

Tool Management

FR-5: Owners shall be able to add new tool listings.

FR-6: Owners shall be able to edit or remove their tool listings.

FR-7: Owners shall be able to specify the tool category, description, condition, availability, and lending agreement.

FR-8: The system shall display all available tools to borrowers.

Search and Reservation

FR-9: Borrowers shall be able to search tools using their name, category, and availability.

FR-10: Borrowers shall be able to submit borrowing requests for available tools.

FR-11: Owners shall be able to approve or reject borrowing requests.

FR-12: The system shall prevent multiple reservations for the same tool during overlapping time periods.

FR-13: The system shall maintain the reservation status throughout the borrowing process.

Borrowing and Returns

FR-14: The system shall record borrowing and return dates.

FR-15: Borrowers shall be able to mark borrowed tools as returned.

FR-16: The system shall automatically update the availability status of a returned tool.

Reviews and Notifications

FR-17: Borrowers shall be able to rate and review completed borrowings.

FR-18: The system shall provide in-app notifications for important events such as request approvals, rejections, and return reminders.

Administration

FR-19: Administrators shall be able to manage user accounts.

FR-20: Administrators shall be able to manage tool listings.

FR-21: Administrators shall be able to resolve disputes between users.

FR-22: Administrators shall be able to view basic platform statistics.

3.2 Non-Functional Requirements

Security

NFR-1: The system shall authenticate users using JWT-based authentication.

NFR-2: Users shall only access features permitted by their assigned role.

Performance

NFR-3: The system should respond to user requests within an acceptable time under normal operating conditions.

Reliability

NFR-4: The system shall maintain data consistency for users, tools, and reservations.

Usability

NFR-5: The user interface shall be simple, intuitive, and easy to navigate.

Maintainability

NFR-6: The application shall be modular to support future enhancements and maintenance.

Availability

NFR-7: The system shall be available whenever the hosting environment is operational.

3.3 External Interface Requirements

User Interface

The system shall provide a web-based graphical user interface with the following pages:

- Login
- Registration
- User Dashboard

- Tool Listings
 - Tool Details
 - Reservation History
 - Owner Dashboard
 - Administrator Dashboard
 - User Profile
 - Notifications
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Hardware Interface

The system requires a device capable of running a modern web browser with internet connectivity.

Software Interface

The application shall interface with:

- React (Frontend)
 - Node.js with Express.js (Backend)
 - MySQL (Database)
 - JSON Web Token (JWT) for authentication
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3.4 Business Rules

BR-1: Every user must register before accessing platform services.

BR-2: Only owners can create, modify, or remove their tool listings.

BR-3: A borrower cannot reserve a tool that is already reserved during the requested time period.

BR-4: Every borrowing request must be approved or rejected by the respective owner.

BR-5: Both the owner and borrower must agree to the lending agreement before the borrowing process begins.

BR-6: Reviews may only be submitted after a reservation has been successfully completed.

BR-7: Administrators have authority to manage users, listings, and resolve disputes whenever necessary.